

lecture 20 : examples

19/11/23

- a) find the points of max/min of $f(x,y) = xy$ over the circle $x^2 + y^2 = 1$
- b) find the points of max/min of $f(x,y,z) = y + z$ on the sphere $x^2 + y^2 + z^2 = 1$
- c) find the closest and furthest point from $(0,0)$ on the curve $2x^2 + 3y^2 - 1 = 0$

$$f(x,y) = xy \quad g(x,y) = x^2 + y^2 - 1 = 0$$

we seek $(x,y), \lambda$ s.t.
$$\begin{cases} f_x(x,y) = \lambda g_x(x,y) \\ f_y(x,y) = \lambda g_y(x,y) \\ g(x,y) = 0 \end{cases}$$

so
$$\begin{cases} y = 2\lambda x & \textcircled{1} \\ x = 2\lambda y & \textcircled{2} \\ x^2 + y^2 - 1 = 0 & \textcircled{3} \end{cases}$$

remember: any points you find must satisfy all 3 conditions

sub $\textcircled{1}$ into $\textcircled{2}$: $x = 2\lambda(2\lambda x) = 4\lambda^2 x \Rightarrow x - 4\lambda^2 x = 0$

factorize: $x(1 - 4\lambda^2) = 0 \Rightarrow \boxed{x=0}$ or $1 - 4\lambda^2 = 0 \Rightarrow \lambda^2 = \frac{1}{4}$

if $\underline{x=0}$ $\textcircled{1}$ ^{or} ~~and~~ $\textcircled{2} \Rightarrow y=0$. But does ~~(0,0)~~ satisfy all? $\Rightarrow \lambda = \pm 1/2$
no!

$\textcircled{3}$: $0^2 + 0^2 - 1 = 0 \Rightarrow -1 = 0 \neq$

sub: $\textcircled{2}$ into $\textcircled{1}$: $y = 2\lambda(2\lambda y) \Rightarrow \boxed{y=0}$ or $\boxed{\lambda = \pm 1/2}$

if $\underline{y=0}$ $\textcircled{1}$ or $\textcircled{2} \Rightarrow x=0$ ~~*~~

what do we get if we sub $\boxed{x=0}$ or $\boxed{y=0}$ into $\textcircled{3}$?

$0^2 + y^2 - 1 = 0 \Rightarrow y = \pm 1$. Does ~~(0,1)~~, ~~(0,-1)~~ work?

$$\begin{cases} 1 = 2\lambda \cdot 0 & \text{no} \\ 0 = 2\lambda \cdot 1 & \checkmark \\ 0^2 + 1^2 - 1 = 0 & \checkmark \end{cases}$$

if $y=0$: $x^2 + 0^2 - 1 = 0 \Rightarrow x = \pm 1$. But $(1,0), (-1,0)$?
this causes a contradiction in $\textcircled{2}$

$$\text{if } \lambda = \frac{1}{2}$$

$$\begin{cases} y = x \\ x = y \\ x^2 + y^2 - 1 = 0 \end{cases}$$

$$\begin{cases} y = \lambda z x & \textcircled{1} \\ x = \lambda z y & \textcircled{2} \\ x^2 + y^2 - 1 = 0 & \textcircled{3} \end{cases}$$

use $\textcircled{3}$: $x^2 + x^2 - 1 = 0 \Rightarrow 2x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}}$

since $y = x$, $y = \pm \frac{1}{\sqrt{2}}$

$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}})$ satisfy all 3 conditions

can write as $(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}})$

$$\text{if } \lambda = -\frac{1}{2} \quad \begin{cases} y = -x \\ x = -y \\ x^2 + y^2 - 1 = 0 \end{cases}$$

$\textcircled{3} \Rightarrow x^2 + (-x)^2 - 1 = 0 \Rightarrow x^2 + x^2 - 1 = 0 \Rightarrow x = \pm \frac{1}{\sqrt{2}}$

$y = -x$ so $(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}})$ and $(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ satisfy all 3 conditions

you can write these as $(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{\sqrt{2}})$

which points lead to min/max of $f(x,y) = xy$
evaluate:

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$$

$$f\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) = \frac{1}{2}$$

$$f\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) = -\frac{1}{2}$$

$$f\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = -\frac{1}{2}$$

$\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$ are points of maximum

$\left(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{\sqrt{2}}\right)$ are points of minimum

$$f(x, y, z) = y + z$$

$$h(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$$

we seek $(x, y, z), \lambda$ s.t.
$$\begin{cases} f_x = \lambda h_x \\ f_y = \lambda h_y \\ f_z = \lambda h_z \\ h = 0 \end{cases}$$

so
$$\begin{cases} 0 = \lambda 2x & (1) \Rightarrow \lambda \neq 0 \text{ or } \boxed{x=0} \text{ always} \\ 1 = \lambda 2y & (2) \Rightarrow y = \frac{1}{2\lambda} \\ 1 = \lambda 2z & (3) \Rightarrow z = \frac{1}{2\lambda} \\ x^2 + y^2 + z^2 - 1 = 0 & (4) \end{cases} \Rightarrow z = y$$

$$(4) \Rightarrow 0^2 + y^2 + y^2 - 1 = 0 \Rightarrow 2y^2 - 1 = 0 \Rightarrow y = \pm \frac{1}{\sqrt{2}} = z$$

$$(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (0, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}) \text{ satisfy all 4 conditions}$$

in this case, $\lambda = \frac{1}{2y} = \frac{1}{2z} = \frac{1}{2(\pm 1/\sqrt{2})} = \pm \frac{1}{\sqrt{2}} = \pm \frac{1}{\sqrt{2}}$

$$(4) \Rightarrow 0^2 + \left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 - 1 = 0 \Rightarrow \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} - 1 = 0$$

$$\Rightarrow \frac{2}{4\lambda^2} - 1 = 0 \Rightarrow \frac{1}{2\lambda^2} = 1 \Rightarrow 1 = 2\lambda^2 \Rightarrow \lambda^2 = \frac{1}{2} \Rightarrow \lambda = \pm \frac{1}{\sqrt{2}}$$

$$y = \frac{1}{2\lambda} = \pm \frac{1}{\sqrt{2}}, \quad z = \pm \frac{1}{\sqrt{2}}$$

$$\textcircled{3} \Rightarrow \lambda = \frac{1}{2\bar{z}}$$

$$0 = \lambda z x \quad \textcircled{1}$$

$$1 = \lambda z y \quad \textcircled{2}$$

$$1 = \lambda z z \quad \textcircled{3}$$

sub into $\textcircled{2}$: $1 = \frac{1}{z\bar{z}} z y \Rightarrow 1 = y z$

$$y = \frac{1}{z} \quad (z \neq 0)$$

$$x^2 + y^2 + z^2 - 1 = 0 \quad \textcircled{4}$$

$$\textcircled{4}: 0^2 + \left(\frac{1}{z}\right)^2 + z^2 - 1 = 0 \quad \text{solve for } z?$$

$$\frac{1}{z^2} + z^2 - 1 = 0 \quad \times z^2 \Rightarrow 1 + z^4 - z^2 = 0$$

$$\Rightarrow z^4 - z^2 + 1 = 0 \Rightarrow (z^2)^2 - (z^2) + 1 = 0$$

$$z^2 = \frac{1 \pm \sqrt{1 - 4 \cdot 1 \cdot 1}}{2 \cdot 1} = \frac{1 \pm \sqrt{-3}}{2} \quad \text{not real!}$$

$$f(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} . \quad f = y + z$$

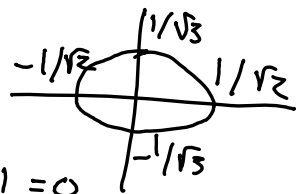
$$f(0, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}) = \frac{-1}{\sqrt{2}} + \frac{-1}{\sqrt{2}} = -\frac{2}{\sqrt{2}} = -\sqrt{2}$$

$(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ is the point of maximum

$(0, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}})$ is the point of minimum

find the points nearest and furthest from $(0,0)$
on the curve $2x^2 + 3y^2 - 1 = 0 \Rightarrow \frac{x^2}{1/2} + \frac{y^2}{1/3} = 1$

what are we optimizing? distance
optimize $\sqrt{x^2 + y^2}$



what is the constraint? $g(x,y) = 2x^2 + 3y^2 - 1 = 0$

we can equivalently optimize $\boxed{x^2 + y^2}$

square-root function is constantly increasing (monotonic)

$$\boxed{f(x,y) = x^2 + y^2}$$

we seek $(x,y), \lambda$ s.t.

$$\begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ g(x,y) = 0 \end{cases}$$

so

$$\begin{cases} 2x = \lambda 4x & (1) \\ 2y = \lambda 6y & (2) \\ 2x^2 + 3y^2 - 1 = 0 & (3) \end{cases}$$

$$\begin{cases} 2x = \lambda 4x & (1) \Rightarrow 2x - 4\lambda x = 0 \Rightarrow x(2 - 4\lambda) = 0 \\ 2y = \lambda 6y & (2) \Rightarrow 2y - 6\lambda y = 0 \Rightarrow y(2 - 6\lambda) = 0 \\ 2x^2 + 3y^2 - 1 = 0 & (3) \end{cases}$$

$$(1) \Rightarrow \boxed{x=0} \text{ or } \boxed{\lambda = \frac{1}{2}} \text{ if } \lambda = \frac{1}{2} \cdot (2) \Rightarrow 2y = \frac{1}{2} \cdot 6y = 3y$$

$$(2) \Rightarrow \boxed{y=0} \text{ or } \boxed{\lambda = \frac{1}{3}} \Rightarrow y=0$$

$$\text{if } \lambda = \frac{1}{3} (1) \Rightarrow 2x = 4 \cdot \frac{1}{3} x \Rightarrow x = 0$$

we can see that $(0,0)$ will not satisfy (3).

• if $x=0$. use (3): $2 \cdot 0^2 + 3y^2 - 1 = 0 \Rightarrow 3y^2 - 1 = 0$

$$\Rightarrow y^2 = \frac{1}{3} \text{ so } y = \pm \frac{1}{\sqrt{3}} \text{ so } (0, \frac{1}{\sqrt{3}}), (0, \frac{-1}{\sqrt{3}})$$

satisfy? (1)✓ (3)✓✓ (2)? $2 \cdot \frac{1}{\sqrt{3}} = \lambda \cdot 6 \cdot \frac{1}{\sqrt{3}} \Rightarrow \lambda = \frac{1}{3}$ nothing new

$(0, \pm \frac{1}{\sqrt{3}})$ satisfy all 3 conditions

• if $y=0$: use (3): $2x^2 + 3(0)^2 - 1 = 0 \Rightarrow 2x^2 - 1 = 0$

$$\Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}} \text{ so } (\frac{1}{\sqrt{2}}, 0), (\frac{-1}{\sqrt{2}}, 0)$$

satisfy? (3)✓ (2)✓ (1)? $2(\frac{1}{\sqrt{2}}) = \lambda 4(\frac{1}{\sqrt{2}}) \Rightarrow \lambda = \frac{1}{2}$ fine

$(\pm \frac{1}{\sqrt{2}}, 0)$ satisfy all 3 conditions.

$$f(x,y) = x^2 + y^2$$

check $f^*(x,y) = \sqrt{x^2 + y^2}$

If you were asked for value of
max/min distance, find pts
with $x^2 + y^2$, determine distance
with $\sqrt{x^2 + y^2}$)

in this question, f^* is not
necessary

$$f(x, y) = x^2 + y^2$$

$$f^*(x, y) = \sqrt{x^2 + y^2}$$

$$f\left(0, \frac{1}{\sqrt{3}}\right) = 0^2 + \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

$$\sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$$

$$f\left(0, -\frac{1}{\sqrt{3}}\right) = \frac{1}{3}$$

$$f\left(\frac{1}{\sqrt{2}}, 0\right) = \left(\frac{1}{\sqrt{2}}\right)^2 + 0^2 = \frac{1}{2}$$

$$\sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

$$f\left(-\frac{1}{\sqrt{2}}, 0\right) = \frac{1}{2}$$

$$\text{still: } \frac{1}{\sqrt{2}} > \frac{1}{\sqrt{3}}$$

$\frac{1}{2} > \frac{1}{3}$ so $\left(\pm \frac{1}{\sqrt{2}}, 0\right)$ are pts of maximum

$\left(\pm \frac{1}{\sqrt{3}}, 0\right)$ are pts of minimum